

Aircraft positioning model

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1 Sets

- $\mathcal{S}$: Slots
- $\mathcal{J}$: Jobs
- $\mathcal{R}$: Airplanes
- $\mathcal{J}_r$: Jobs corresponding to plane $r \in \mathcal{R}$
- $\mathcal{P}$: Positions

2 Parameters

- H : Time horizon
- D_j : Duration of job $j \in \mathcal{J}$
- $B_{pp'}$: 1 if positions $p \in \mathcal{P}$ and $p' \in \mathcal{P}$ interfere with each other, 0 otherwise
- $P_{jj'}$: 1 if job $j \in \mathcal{J}$ immediately precedes $j' \in \mathcal{J}$, 0 otherwise
- $A_{jj'}$: 1 if job $j \in \mathcal{J}$ and job $j' \in \mathcal{J}$ correspond to the same plane, 0 otherwise
- R_j : Airplane of job $j \in \mathcal{J}$

3 Variables

x_{spj}	:	1 if job $j \in \mathcal{J}$ is performed in slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$
y_{spr}	:	1 if airplane $r \in \mathcal{R}$ is in slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$
z_{sp}	:	1 if job in slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$ is different from the job in slot $s + 1 \in \mathcal{S}$
w_{rp}	:	1 if some job of plane $r \in \mathcal{R}$ is performed in position $p \in \mathcal{P}$
d_{sp}^S	:	duration of slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$
d_{spj}^J	:	duration of slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$ for job $j \in \mathcal{J}$
s_{spj}^J	:	starting time of slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$ for job $j \in \mathcal{J}$
f_{spj}^J	:	finishing time of slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$ for job $j \in \mathcal{J}$
s_{spj}^S	:	starting time of slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$
f_{spj}^S	:	finishing time of slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$
$\alpha_{ss'pp'}$	:	slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$ interferes with slot $s' \in \mathcal{S}$ of position $p' \in \mathcal{P}$ if $B_{pp'} = 1$. This means that either the starting or the finishing times of slot s are in between the starting and the finishing times of slot s'
$\beta_{ss'pp'}^S$	:	1 if starting time of slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$ is earlier than the starting time of slot $s' \in \mathcal{S}$ of position $p' \in \mathcal{P}$ if $B_{pp'} = 1$, 0 otherwise
$\beta_{ss'pp'}^F$	:	1 if finishing time of slot $s \in \mathcal{S}$ of position $p \in \mathcal{P}$ is later than the finishing time of slot $s' \in \mathcal{S}$ of position $p' \in \mathcal{P}$ if $B_{pp'} = 1$, 0 otherwise

4 Constraints

Each slot of each position can have one job at a time:

$$\sum_{j \in \mathcal{J}} x_{spj} \leq 1 \quad \forall s \in \mathcal{S}, \forall p \in \mathcal{P} \quad (1)$$

Airplane-job consistency assignment

$$y_{spr} = \sum_{j \in \mathcal{J}_r} x_{spj} \quad \forall s \in \mathcal{S}, \forall p \in \mathcal{P}, \forall r \in \mathcal{R} \quad (2)$$

Airplane with some job in a position.

$$w_{rp} \geq y_{spr} \quad \forall s \in \mathcal{S}, \forall p \in \mathcal{P}, \forall r \in \mathcal{R} \quad (3)$$

Calculation of the slot duration

$$d_{spj}^J = f_{spj}^J - s_{spj}^J \quad \forall s \in \mathcal{S}, \forall p \in \mathcal{P}, \forall j \in \mathcal{J} \quad (4)$$

Starting and finishing times are 0 if the job is not assigned to a position

$$\begin{aligned} s_{spj}^J &\leq H \times x_{spj} \quad \forall s \in \mathcal{S}, \forall p \in \mathcal{P}, \forall j \in \mathcal{J} \\ f_{spj}^J &\leq H \times x_{spj} \quad \forall s \in \mathcal{S}, \forall p \in \mathcal{P}, \forall j \in \mathcal{J} \end{aligned} \quad (5)$$

The total duration of a job is the sum of the duration of all corresponding slots:

$$\sum_{s \in \mathcal{S}} \sum_{p \in \mathcal{P}} d_{spj}^J = D_j, \forall j \in \mathcal{J} \quad (6)$$

The duration of a slot is that of the slot assigned to that job

$$d_{sp}^S = \sum_{j \in \mathcal{J}} d_{spj}^J, \forall s \in \mathcal{S}, \forall p \in \mathcal{P} \quad (7)$$

The starting and finishing times of a slot are those of the slot assigned to that job

$$\begin{aligned} s_{sp}^S &= \sum_{j \in \mathcal{J}} s_{spj}^J, \forall s \in \mathcal{S}, \forall p \in \mathcal{P} \\ f_{sp}^S &= \sum_{j \in \mathcal{J}} f_{spj}^J, \forall s \in \mathcal{S}, \forall p \in \mathcal{P} \end{aligned} \quad (8)$$

Computing if the starting time of slot s in position p is earlier than the starting time of slot s' in position p' :

$$H \times \beta_{s,s',p,p'}^S + s_{sp}^S \geq s_{s'p'}^S, \forall s, s' \in \mathcal{S}, \forall p, p' \in \mathcal{P} / B_{pp'} = 1 \quad (9)$$

Computing if the finishing time of slot s in position p is later than the starting time of slot s' in position p'

$$H \times \beta_{s,s',p,p'}^F + s_{sp}^S \geq f_{s'p'}^S, \forall s, s' \in \mathcal{S}, \forall p, p' \in \mathcal{P} / B_{pp'} = 1 \quad (10)$$

The starting time of slot s in position p is earlier than the starting time of slot s' in position p' and finishing time of slot s in position p is later than the starting time of slot s' in position p'

$$1 + \alpha_{ss'pp'} \geq \beta_{ss'pp'}^S + \beta_{ss'pp'}^F, \forall s \in \mathcal{S}, s' \in \mathcal{S}, \forall p \in \mathcal{P}, \forall p' \in \mathcal{P} / B_{pp'} = 1 \quad (11)$$

Slot within the same position are sequenced:

$$s_{sp}^S \leq f_{s-1p}^S, \forall s \in \mathcal{S} \setminus s = 1, p \in \mathcal{P} \quad (12)$$

Jobs are sequenced:

$$\sum_{s \in \mathcal{S}} \sum_{p \in \mathcal{P}} s_{spj}^J \leq \sum_{s \in \mathcal{S}} \sum_{p \in \mathcal{P}} f_{spj'}^J, \forall j, j' \in \mathcal{J} / P_{jj'} = 1 \quad (13)$$

Subsequent tasks in a position correspond to different planes:

$$1 + z_{sp} \geq y_{spr} + y_{s+1pr'}, \forall s \in \mathcal{S}, \forall p \in \mathcal{P}, \forall r, r' \in \mathcal{R} / r \neq r' \quad (14)$$

5 Objective function

$$\min. \sum_{s \in \mathcal{S}} \sum_{p \in \mathcal{P}} \sum_{j \in \mathcal{J}} x_{spj} \quad (15)$$

$$+ \sum_{s \in \mathcal{S}} \sum_{s' \in \mathcal{S}} \sum_{p \in \mathcal{P}} \sum_{p' \in \mathcal{P}} \alpha_{ss'pp'} \quad (16)$$

$$+ \sum_{s \in \mathcal{S}} \sum_{p \in \mathcal{P}} \sum_{r \in \mathcal{R}} y_{spr} \quad (17)$$